

SAMPLE SPECIFICATION WITH CHOICES FOR PIPE, VALVES & FITTINGS
For
SDR 13.5 PE IRRIGATION SYSTEMS

PE PIPE:

- A. Polyethylene pipe shall comply with ASTM D3035-14a or ASTM F714-13. Pipe shall be IPS diameter, Standard Dimension Ratio (SDR) 13.5 and made from PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black.
- B. Butt fusion joints shall be made between parts of the same diameter and the same SDR. Unlike SDR butt fusions are not permitted. SDR 11 to SDR 13.5 butt fusions are not permitted.
- C. Only pipe in straight lengths shall be used. Coiled pipe received in rolls is not permitted.

FITTINGS FOR PE PIPE:

- A. Molded butt fusion fittings shall comply with ASTM D3261-12^ε. Fittings shall be IPS diameter, SDR 11, with ends machined to 13.5. They shall be made from PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black. The injection point on the fitting (otherwise known as the gate or sprue) shall be oriented to inject perpendicular to the plane of the fitting and positioned at the center of the fitting so its knit line (weld line) is not located in or near the "crotch" of the fitting. Fittings shall be as manufactured by The Harrington Corporation.
- B. Socket fusion fittings are not permitted.
- C. Fabricated butt fusion fittings shall be made from DR 11 pipe complying with ASTM D3035-14a or ASTM F714-13 with ends machined to DR 13.5. Pipe shall be IPS diameter and made from PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black. Fabricated fittings shall comply with AWWA C906-07. Bends greater than 45⁰ shall have no fewer than 3 segments. Size x Size Tees shall be 3 segment. Reducing tees

whether reducing on branch or run shall have their “main bodies” made from pipe of diameter that is the same as the largest diameter connection.

- D. Electrofusion Fittings shall comply with ASTM F1055. Fittings shall be IPS diameter, and made from PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black. Couplings, saddles, and fittings shall be SDR 11 or better. Branches of branch saddles shall be machined to SDR 13.5 when to be butt fused to an SDR 13.5 pipe or fitting. Threaded FNPT service saddles shall have fusion bond epoxy coated ductile iron adapter outlets. NPT threads shall comply with ANSI B1.20.1. Electrofusion x Swivel saddles shall have a fusion bond epoxy coated ductile iron female swivel adapter outlet compatible with Harco male swivel fittings and valves.
- E. Flange adapters shall comply with ASTM F2880-14. Fittings shall be IPS diameter, SDR 11 with ends machined to SDR 13.5, and made from PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black.
- F. Flange Adapter Back Up Rings shall be Ductile Iron per ASTM A536 and SDR 11.
- G. Flange Accessories shall be 1/8” Neoprene Gaskets and Grade 5 or stronger, zinc plated Cap Screws or Threaded Rod and Nuts.
- H. Threaded Transitions: PE x MNPT Brass transitions shall be SDR 11 with end machined SDR 13.5. The threaded end shall be Red Brass. NPT threads shall comply with ANSI B1.20.1. The material shall be PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black.
- I. Mechanical Joint Adapters (male or female) shall comply with AWWA C906-07 and be SDR 11 with ends machined SDR 13.5. The MJ end shall employ a stainless steel stiffener. Material shall be PE4710 compound meeting ASTM D3350 Code C with 2% to 3% carbon black
- J. Mechanical Joint Accessories shall use ductile iron glands, T-Bolts/nuts, and gaskets per AWWA C111. Joint restraint components required for female adapters shall be compatible with the connection being made.

- K. Stiffeners shall be used in PE ends when installed in gasketed joints. Stiffeners shall be 316 Stainless Steel and sized to fit the diameter and SDR of the end it is used in.
- L. Compression fittings shall be suitable for use on IPS diameter PE pipe per ASTM D3035-14a or ASTM F714-13. Fittings shall be long term rated for 230 psi complying with ISO 14236 and meet the dimensional and performance requirements of AWWA C800. Fitting “bodies” shall be polypropylene. Grip rings shall be acetal. Nuts shall be polypropylene or acetal. Seals shall be nitrile. Joint seal and restraint activation shall be accomplished solely by the compression nut. Joint seals shall not interfere with pipe insertion. No bevelling or lubrication of pipe shall be required. NPT threads shall comply with ANSI B1.20.1. Acme threads shall comply with ASTM F2768. Compression fittings shall be manufactured by Phimac and supplied by The Harrington Corporation.
- M. Side wall fusion saddles are not permitted.

ISOLATION VALVES:

- A. Iron Gate Valves shall comply with AWWA C509 and have flanged ends. Iron gate valves shall be as manufactured by Clow or Nibco.
- B. Angle Valves shall be “full port” and 250 psi rated with integrally restrained ends. Inlet end shall be male swivel and outlet end shall be integrally restrained push on gasketed joint with “floating grip ring” and a single clamp bolt. Inlet end shall provide orientation of the outlet independent of joint seal without the use of additional components. Components with wetted surfaces, all bolts & nuts, and cross handles shall be 316 Stainless Steel. The seal assembly shall retract fully from the valve body into the valve bonnet and out of the flow stream. Grip rings shall be ductile iron with machined serrations. Gasketed joints shall be SBR rubber and comply with ASTM F477. Internal seals shall be peroxide cured EPDM rubber. There shall be two stem seals. Angle Valves shall be as manufactured by The Harrington Corporation.